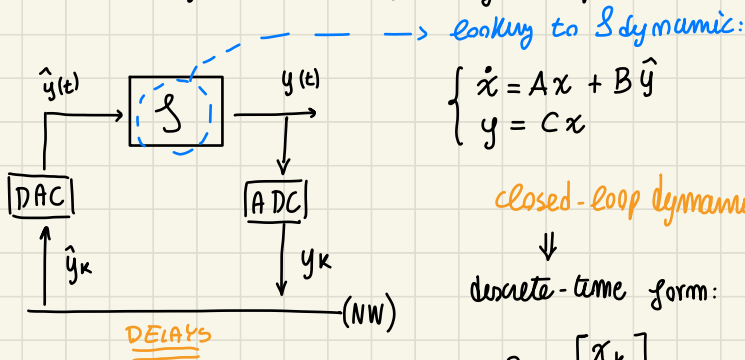


30/11

we considered a system $\mathcal{S}$: with following set-up

$$z_k = \begin{bmatrix} x_k \\ \hat{y}_k \end{bmatrix}$$

expanding the states in this form, we obtain:

$$z_{k+1} = \Phi(h_k, \tau_k) z_k \quad \left(\begin{array}{l} \text{in time varying parameters} \\ h_k, \tau_k \end{array} \right)$$

$$\Phi(h_k, \tau_k) = \begin{bmatrix} e^{A h_k} + \Gamma(h_k - \tau_k) B C & e^{A(h_k - \tau_k)} \Gamma(\tau_k) B \\ C & 0 \end{bmatrix}$$

where

$$\Gamma(\mu) = \int_0^\mu e^{A\nu} d\nu$$

~~LTI~~ $\Rightarrow$ LTV or LPV (Linear Parameter Varying)

dynamic depends on parameters
changing in time

interesting to study dynamic under constant parameter assumption...

How to study stability of time varying syst ?!

(CASE 1) $h_k = h, \tau_k = \tau \quad (h > \tau \geq 0)$

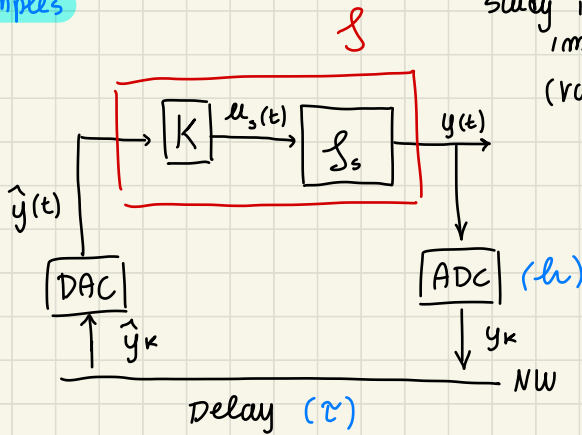
$$z_{k+1} = \bar{\Phi}(h, \tau) z_k$$

LTI $\rightarrow$ easy to study stability $\Rightarrow$ the Network control system (NCS) is asympt stable IFF $\bar{\Phi}(h, \tau)$ is schur stableall eig of it s.t. $|eig(\cdot)| < 1 \Rightarrow$

i.e. $\exists P > 0$ s.t. $\Phi^T(h, \tau) P \Phi(h, \tau) - P < 0$ (Lyapunov)

(even when removing constant assumption.. similar stability condition!)

examples



study in simple cases, the impact on the choice of K !
(range of K which guarantee stability vs. in digital control)

under static output-feedback control

with

$$\begin{cases} \dot{x}_s = A_s x_s + B_s u_s \\ y = C_s x_s \end{cases}$$

(how to write $\Phi(h, \tau)$ as function of K)

In that scheme $u_s(t) = K \hat{y}(t)$

$$\begin{cases} \dot{x}_s = \overset{A}{\underset{=}{A_s}} x_s + \overset{B}{\underset{=}{B_s}} K \hat{y}(t) \\ y = \overset{C}{\underset{=}{C_s}} x_s \end{cases}$$

---> from here analyze as usual stability property

In this case:

$$\Phi(h, \tau) = \begin{bmatrix} e^{A_s h} + \Gamma(h, \tau) B_s K C_s & e^{A_s (h-\tau)} \Gamma(\tau) B_s K \\ C_s & 0 \end{bmatrix}$$

in general NOT easy to check K role on syst stability!

easy when $C_s = I$ (identity)

If $C_s = I \Rightarrow \underbrace{\Phi(h, \tau) = \bar{\Phi}^0 + G^0 K}_{\downarrow \text{to design } K \text{ to give stability!}}$

(transmit $y = x_s$)

↓ IN SCALAR CASE: easy to analyze stability wrt K

scalar system

example

$\dot{x}_s = u$ (integrator)

K that guarantee stability? $\begin{cases} \text{in continuous (1)} \\ \text{in digital (2)} \\ \text{in network (3)} \end{cases}$

(1) continuous time: $u(t) = K \cdot x_s(t)$

$\dot{x}_s = \boxed{K} x_s$ eigenvalues: A.S $\Leftrightarrow \boxed{K < 0}$

(2) digital control

(introduce sampling) $\Rightarrow h$: sample time

1. discretize dyn: $x_{k+1} = f x_k + g u_k$

2. apply control digitally $u_k = K x_k$

$\begin{cases} f = e^{Ah} = 1 \quad (A=0) \\ g = \int_0^h e^{A\gamma} d\gamma B = h \quad (B=1) \end{cases}$

$\int_0^h 1 = h$

$x_{k+1} = x_k + h u_k$ (discrete time integrator)

$= x_k + h K x_k = \boxed{(1 + h K)} x_k$

shun
stability

A.S $\Leftrightarrow |1 + h K| < 1$

$\begin{cases} 1 + h K < 1 \Leftrightarrow h K < 0 \Rightarrow K < 0 \\ -1 < 1 + h K \Leftrightarrow K > -\frac{2}{h} \end{cases}$

$\boxed{-\frac{2}{h} < K < 0}$

$\hookrightarrow (K < 0)$
 $h \rightarrow 0$

as $h \downarrow$, so we get closer to cont time $-\frac{2}{h} \rightarrow -\infty$

same as cont. time (necessary + conservative to cont. time)

↓
(continuous continuous condition + additional same)

condition for stability of network control, where also τ affect? (counter intuitive)

(3) network control system

explicit $\Phi(h, \tau)$ term by term

$$\begin{cases} A_s = 0 \Rightarrow e^{A_s h} = 1 \quad (\text{also } e^{A_s(h-\tau)} = 1) \\ B_s = 1 \\ C_s = 1 \end{cases} \quad \Gamma(\mu) = \int_0^\mu e^{A_s v} dv = \mu$$

$$\Downarrow$$

$$\Phi(h, \tau) = \begin{bmatrix} 1 + (h - \tau)K & \tau K \\ 1 & 0 \end{bmatrix}$$

OK, good theory consistent, what for $\tau > 0$ (delay)

$\tau K \neq 0$ No more triangular matrix Φ

$\downarrow$ usual procedure, compute char poly and study it
OR use JURY CRITERION for 2×2 matrix

system stable $\Leftrightarrow A.S$

for consistency to discrete time

IF $\tau = 0$ (JUST digital control case)

$$\Phi(h, 0) = \begin{bmatrix} 1 + hK & 0 \\ 1 & 0 \end{bmatrix}$$

2×2 matrix, triangular...

with eigenvalues $\lambda_1 = 1 + hK$

same eig of digital control! $\lambda_2 = 0$ A.S! $\in (-1, 1)$

CONSISTENT $\checkmark$

$$p(\lambda) = \det(\lambda I - \phi(h, \tau)) = \det \left(\begin{bmatrix} \lambda - (1 + (h - \tau)K) & -K\tau \\ -1 & \lambda \end{bmatrix} \right) =$$

$$= \boxed{\lambda^2 - (1 + (h - \tau)K)\lambda - K\tau}$$

by JURY CRITERION $p(\lambda) = \lambda^2 + a\lambda + b \rightarrow$ in discrete time: (condition)

$$\Downarrow \begin{cases} -1 < b < 1 \\ -(1+b) < a < (1+b) \end{cases} \quad \begin{array}{l} \text{from application of JURY CRITERION} \\ \text{to a system of ORDER 2} \end{array}$$

$$\begin{cases} -1 < -K\tau < 1 \Rightarrow -1 < K\tau < 1 \\ \text{"} & \text{"} \end{cases} \quad \begin{array}{l} \text{condition (i)} \\ \text{condition (ii)} \end{array}$$

$$\begin{cases} -1 < K\tau < 1 \\ -(1 - K\tau) < 1 + (h - \tau)K < (1 - K\tau) \end{cases}$$

$$-(1-k\tau) < 1 + (h-\tau)k < 1-k\tau$$

$$1 + h k - \cancel{\tau k} < 1 - \cancel{k\tau} \Rightarrow k(h-\tau) < 0 \Leftrightarrow \boxed{k < 0} \quad (*)$$

same as cont/discrete time!
 condition (ii) $\uparrow$

$$1 + (h-\tau)k > -1 + k\tau$$

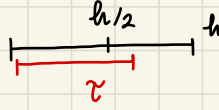
$$\frac{k(h-\tau)}{2} > -2 \Rightarrow \boxed{k\left(\frac{h}{2} - \tau\right) > -1} \quad \text{condition (iii)}$$

NOT TRIVIAL condition..

consider 2 cases $\frac{h}{2} - \tau \geq 0$ according to h, τ relationship,
 we just know $\tau < h$

• IF, (CASE a) $\tau > \frac{h}{2}$ (LARGE DELAY)

$$\begin{cases} \frac{h}{2} - \tau < 0 \\ (*) \quad k < 0 \end{cases}$$



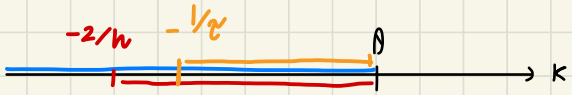
[datum must arrive before h expire!]

$\Downarrow$
 (iii) is always verified
 $\Downarrow$
 in this

$$\begin{cases} k < 0 \\ \frac{h}{2} - \tau < 0 \end{cases} \leadsto k\left(\frac{h}{2} - \tau\right) > 0 \text{ always} \Rightarrow k\left(\frac{h}{2} - \tau\right) > -1!$$

from cond (i)

$$- \frac{1}{\tau} < k < 0$$
 from cond (ii)



• Cont. time, $k < 0$

• Digital control, $-2/h < k < 0$

In case of delay, further condition
 MORE OR LESS conservative than digital?

$$-1/\tau < k < 0 \quad (NCS)$$

more conservative!
 when large delay! be
 more conservative when choosing k , danger, be carefully

$\Leftarrow$ under $\tau > h/2$

due to τ as expected, it make situation more complex...

When optimizing closed loop behaviour (less DOF in K choice)

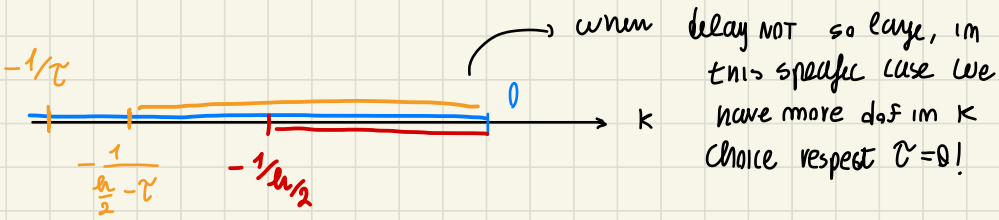
less DOF, less flexibility

$\Leftarrow$ sub-optimality

While, interestingly for small delay, counter intuitive solution !

• Case b $0 \leq \tau < h/2$ (small delay)

$$\left\{ \begin{array}{l} -\frac{1}{\tau} < K < 0 \quad \text{(i) \& (ii) condition} \\ -\frac{1}{\frac{h}{2} - \tau} < K \quad \text{(iii)} \end{array} \right. \quad \begin{array}{l} (\frac{h}{2} - \tau) > 0 \text{ now} \\ \Leftarrow \textcircled{K} (\frac{h}{2} - \tau) > -1 \\ \quad \uparrow \\ \quad K < 0 \end{array}$$



$K < 0$ continuous

$K \in (-1/h/2, 0)$ discrete

$$-\frac{1}{(\frac{h}{2} - \tau)} < -\frac{1}{h/2} \quad \text{for sure}$$

$$\swarrow \text{according to } \tau \left(-\frac{1}{\tau} \geq -\frac{1}{\frac{h}{2} - \tau} \right)$$

Less conservative bound
with respect to Digital control

(because $h/2$ represent equivalent delay induced by sample, When small delay you compensate induced sample delay!)

$\uparrow$ This happens in this simple case! NOT general validity!

THIS IS NOT a RULE

$\Rightarrow$ so, in general τ DOES NOT necessarily means conservative completion!

(CASE (a)) is NOT even a RULE! especially when τ is small

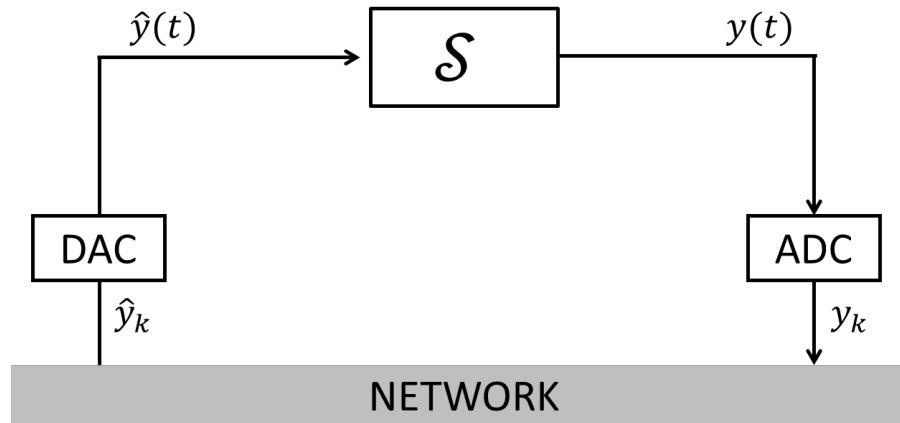
$\hookrightarrow$ let's see some simulations

Outline

- Reference configuration
- **Examples (constant delay)**
- Examples (time-varying sampling time and delay)
- Compensation of delays
- Examples (constant sampling time and delay, with packet dropouts)

Examples: summary

We consider the general NCS scheme



where the dynamics of $\mathcal{S}$ is

$$\begin{cases} \dot{x}(t) &= Ax(t) + B\hat{y}(t) \\ y(t) &= Cx(t) \end{cases}$$

Under Assumption 1 (constant sampling time and delay), the dynamics of the NCS is described by

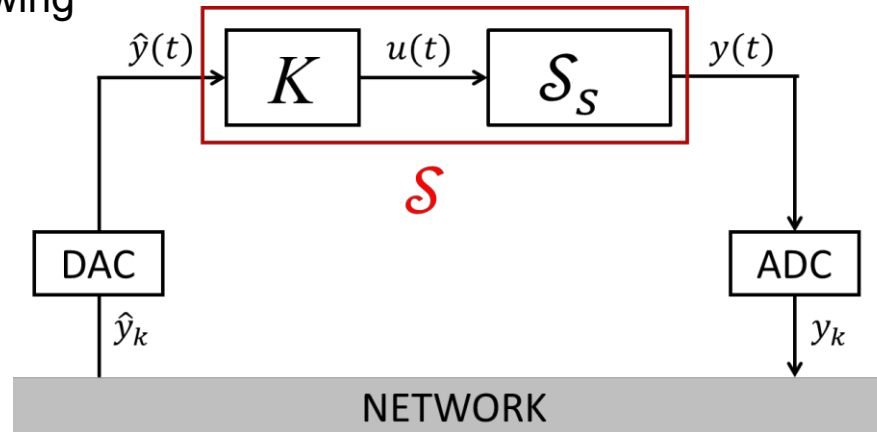
$$z_{k+1} = \Phi(h, \tau)z_k$$

where $z_k = [x_k, \hat{y}_{k-1}]^T$ and

$$\Phi(h, \tau) = \begin{bmatrix} e^{Ah} + \Gamma(h - \tau)BC & e^{A(h-\tau)}\Gamma(\tau)B \\ C & 0 \end{bmatrix}, \Gamma(\xi) = \int_0^\xi e^{Av} dv$$

Examples: summary

In case system $\mathcal{S}$ consists of system+output feedback controller co-located with the actuator, the scheme is the following



where the dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. Renaming $x(t) = x_S(t)$, $A = A_S$, $B = B_S K$, $C = C_S$, we retrieve the scheme of the previous slide. So, we rewrite

$$\Phi(h, \tau) = \begin{bmatrix} e^{A_S h} + \Gamma(h - \tau) B_S K C_S & e^{A_S(h-\tau)} \Gamma(\tau) B_S K \\ C_S & 0 \end{bmatrix}, \Gamma(\xi) = \int_0^\xi e^{A_S v} dv$$

Example 1

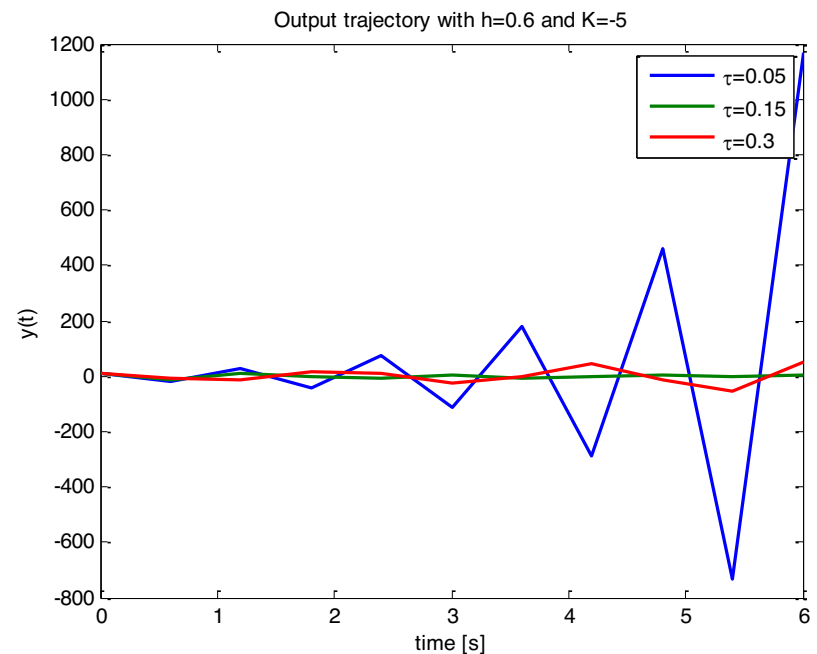
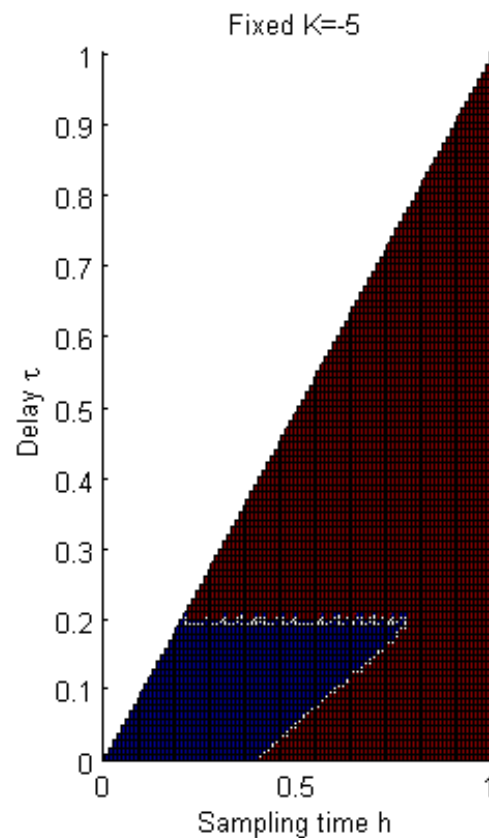
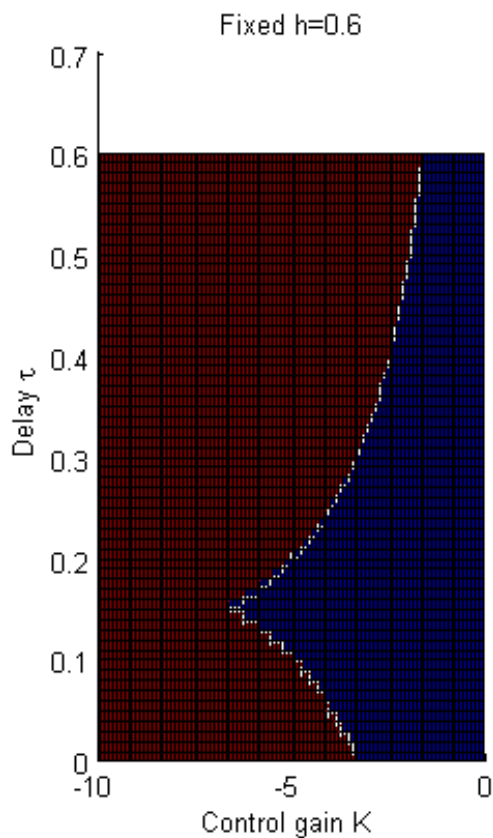
↓ let's look at some simulations!

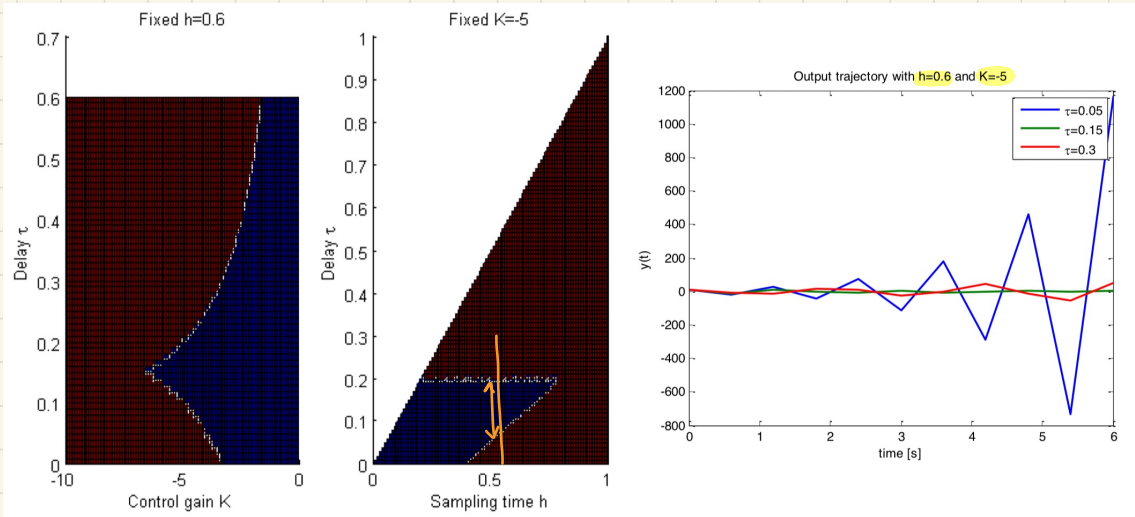
The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= u(t) \\ y(t) &= x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} 1 + K(h - \tau) & K\tau \\ 1 & 0 \end{bmatrix}$$





for $\tau \uparrow$ wider range
 than becomes conservative
 $\downarrow$
 up to τ to 0.2 becomes
 larger, than smaller...

for τ small $\leadsto$ unstable!
 than range of stability

assume τ inside certain range...
 usually I avoid to work on this h
 range... to chose sample time h
 I work where always stability holds

$\tau=0.3$ again unstability slow-up
 $\downarrow$
 IF τ smaller than expected $\tau=0.05$
 BAD!

Example 2

The dynamics of the scalar system $\mathcal{S}_s$ is

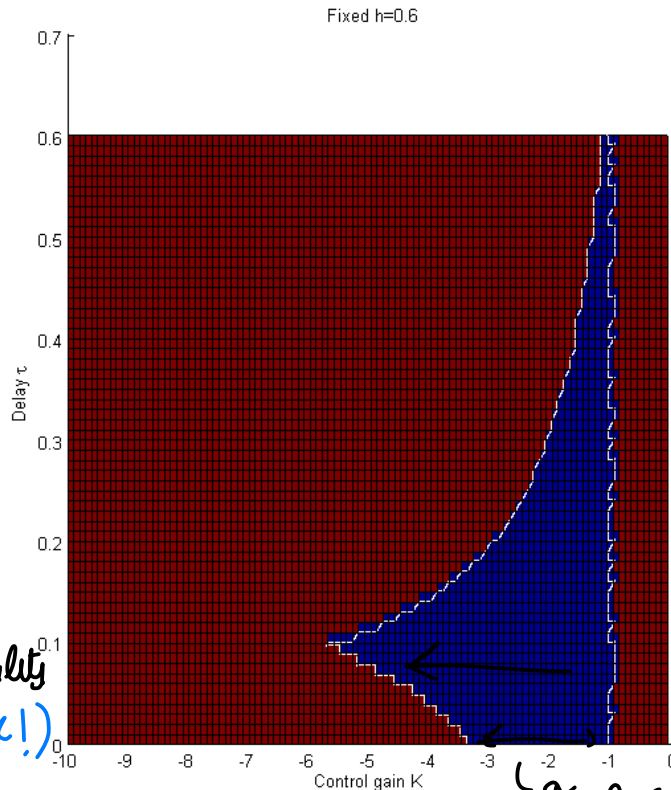
$$\begin{cases} \dot{x}_s(t) = x_s(t) + u(t) \\ y(t) = x_s(t) \end{cases}$$

unstable system dynamic
 $\downarrow$ now new $\tilde{\Phi}(h, \tau)$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

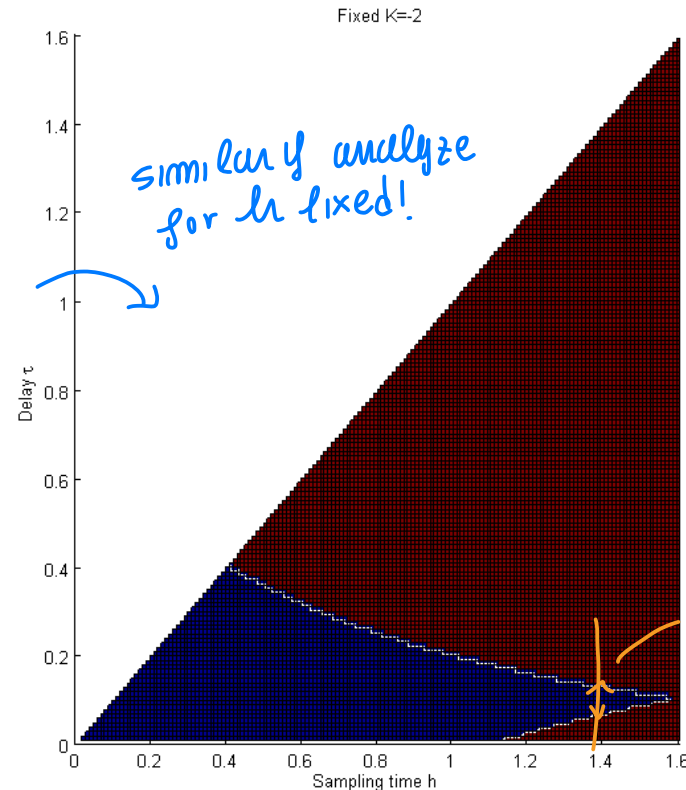
qualitatively similar result!

$$\Phi(h, \tau) = \begin{bmatrix} e^h + K(e^{h-\tau} - 1) & Ke^{h-\tau}(e^\tau - 1) \\ 1 & 0 \end{bmatrix}$$



*but $\tau > 0.1$
worst stability region
 $\uparrow$
 $\tau \in (0, 0.1)$
bigger range as K stability (small τ ok!)*

$\tau = 0$, certain range



similarly analyze for h fixed!

ambiguous case as before

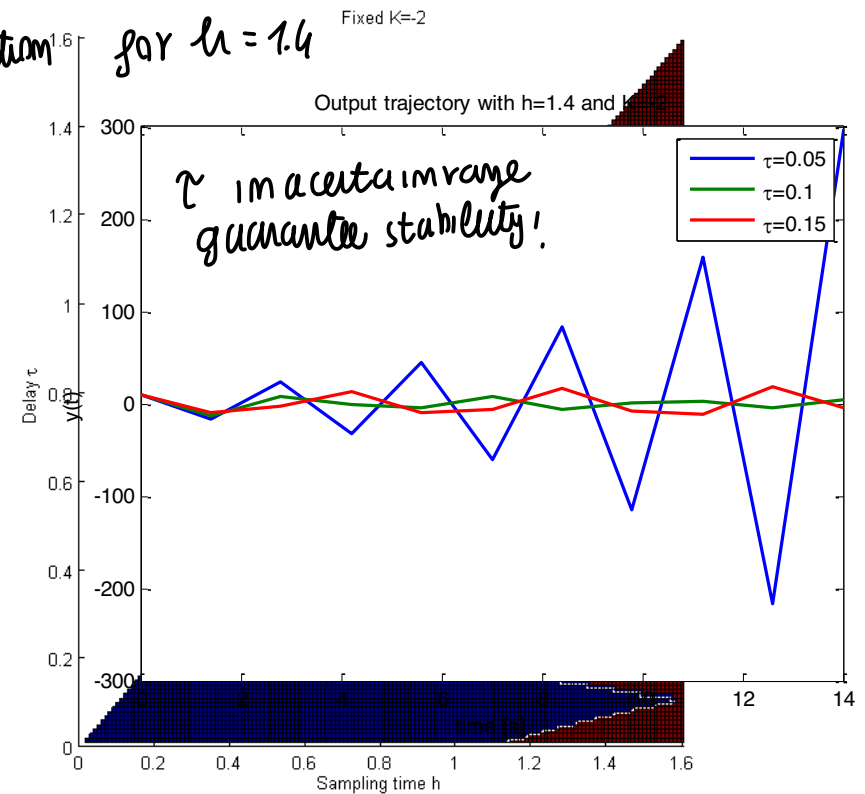
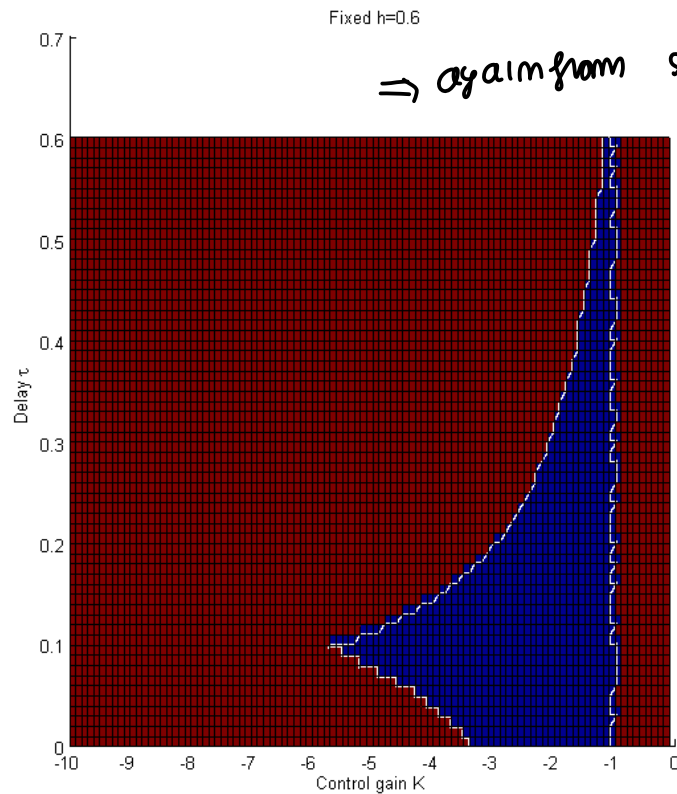
Example 2

The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= x_S(t) + u(t) \\ y(t) &= x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^h + K(e^{h-\tau} - 1) & Ke^{h-\tau}(e^\tau - 1) \\ 1 & 0 \end{bmatrix}$$



Example 3

we have same result for a STABLE open loop syst!

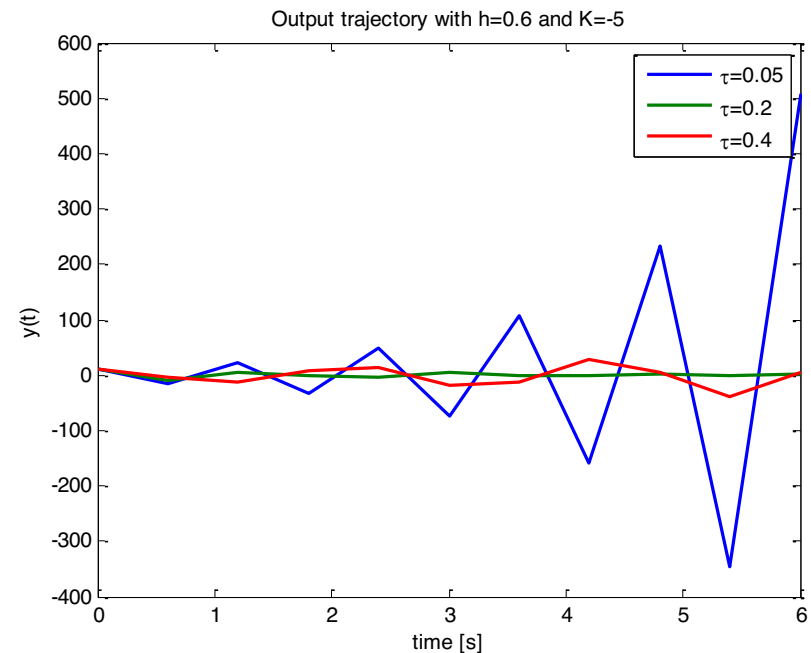
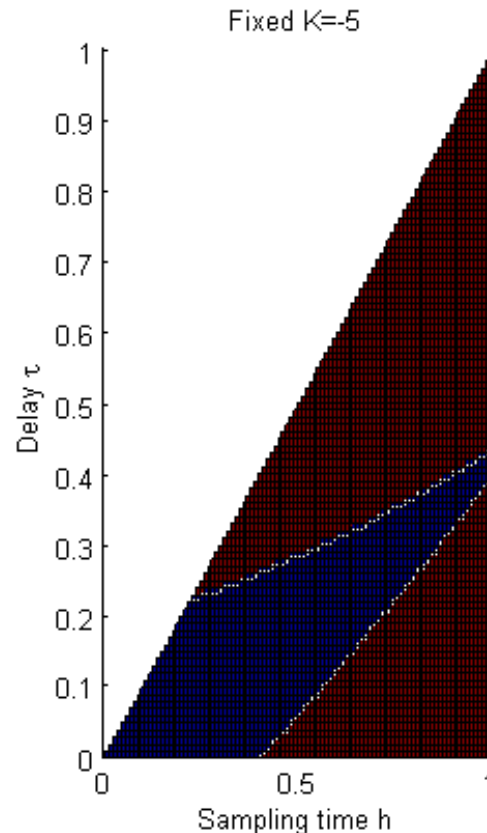
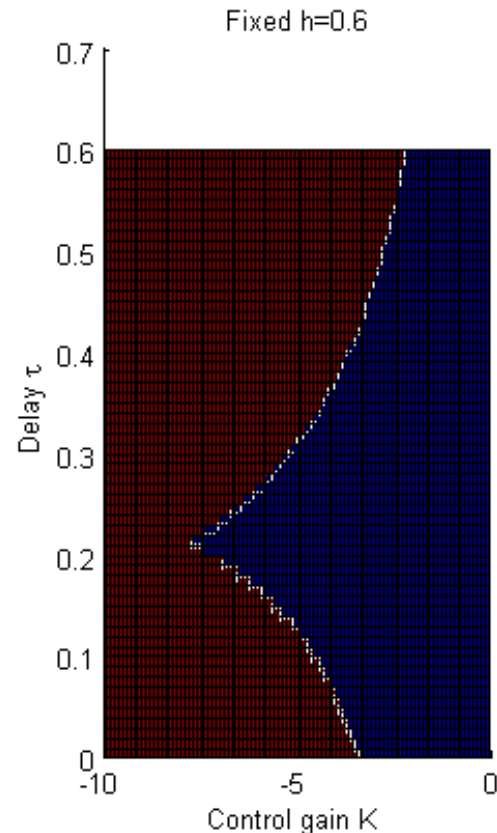
The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) \\ y(t) \end{cases} = \begin{cases} -x_S(t) + u(t) \\ = x_S(t) \end{cases}$$

same for asympt. stable system

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^{-h} + K(1 - e^{-(h-\tau)}) & Ke^{-(h-\tau)}(1 - e^{-\tau}) \\ 1 & 0 \end{bmatrix}$$



Example 4

← that was a scalar system property (this can occur that $\tau \uparrow$ better up to certain range)
 ↓ different from 2x2

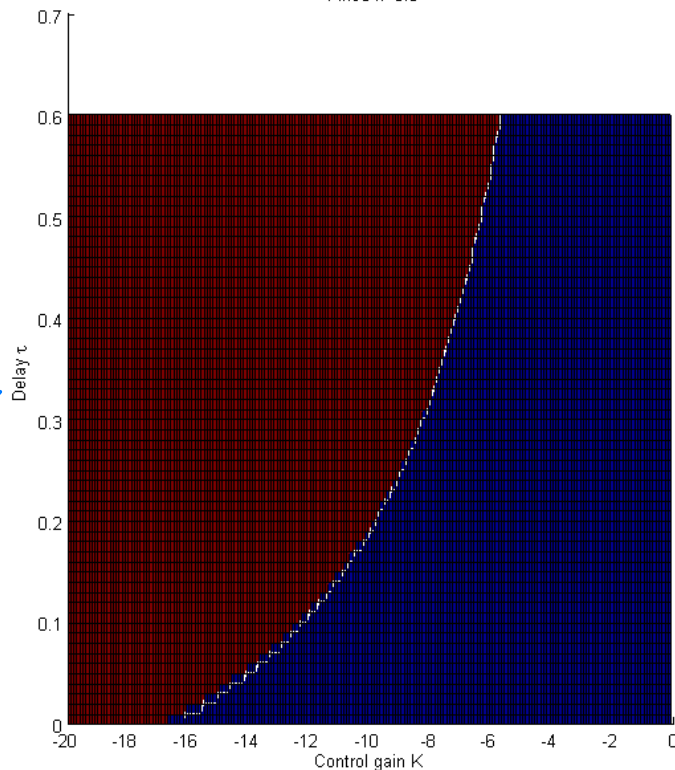
The dynamics of the 2x2 system S_s is

$$\begin{cases} \dot{x}_s(t) = A_s x_s(t) + B_s u(t) \\ y(t) = C_s x_s(t) \end{cases}, A_s = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_s = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_s = \begin{bmatrix} -0.1 & 1 \end{bmatrix}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

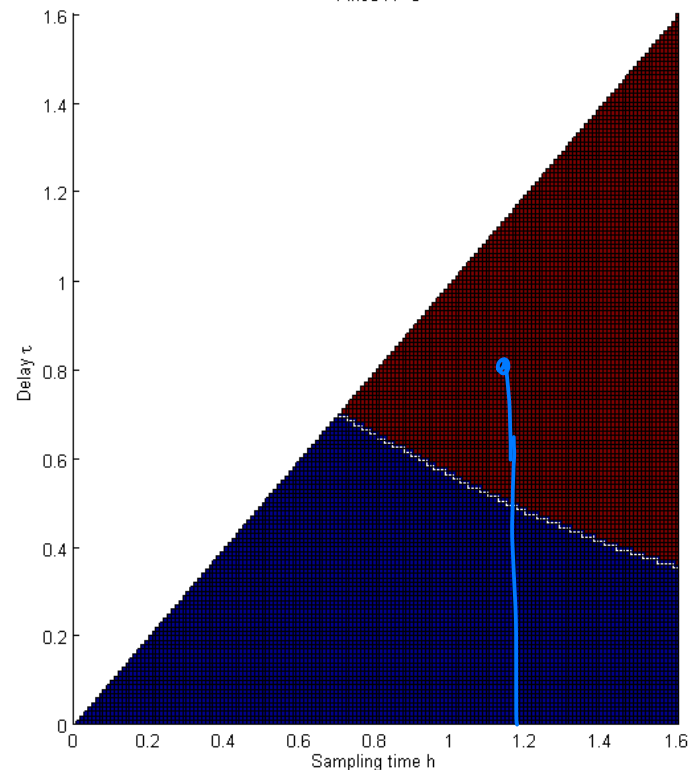
$$\Phi(h, \tau) = \begin{bmatrix} e^{A_s h} + A_s^{-1}(e^{A_s(h-\tau)} - I)B_s K C_s & e^{A_s(h-\tau)} A_s^{-1}(e^{A_s \tau} - I)B_s K \\ C_s & 0 \end{bmatrix}$$

Fixed $h=0.6$



$\tau \uparrow$ decrease range of K stable!

Fixed $K=-5$



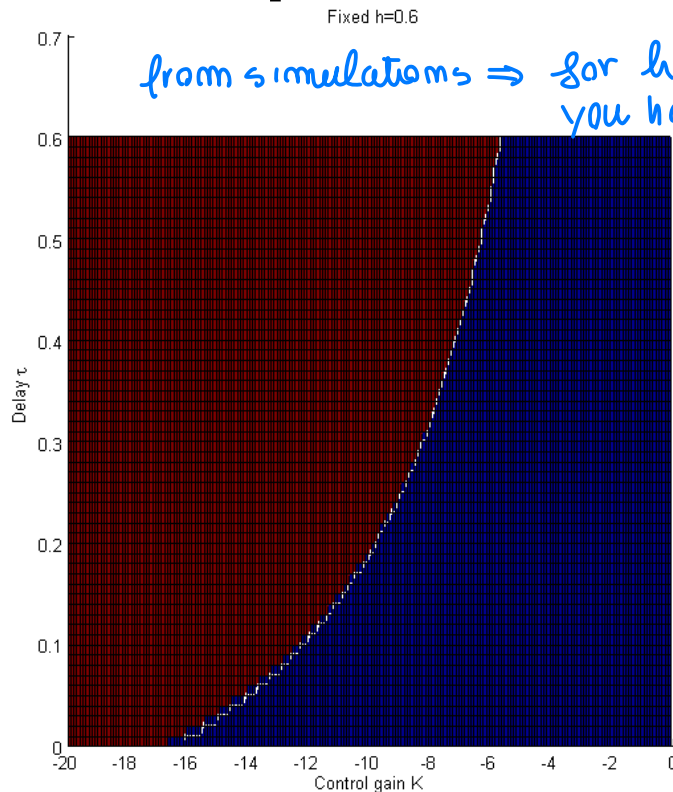
Example 4

The dynamics of the 2x2 system $\mathcal{S}_s$ is

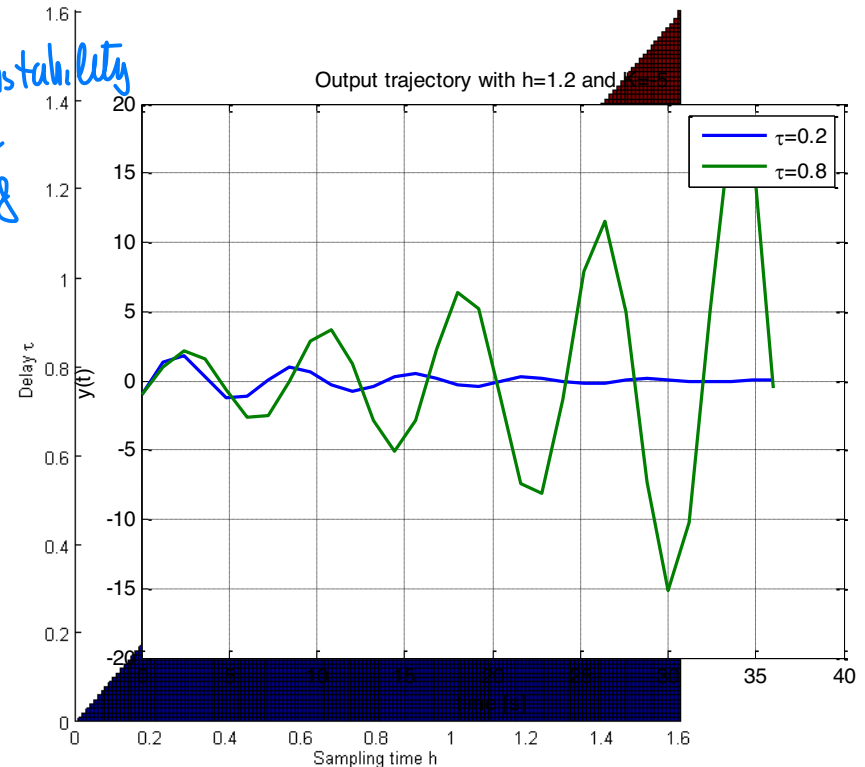
$$\begin{cases} \dot{x}_s(t) = A_s x_s(t) + B_s u(t) \\ y(t) = C_s x_s(t) \end{cases}, A_s = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_s = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_s = \begin{bmatrix} -0.1 & 1 \end{bmatrix}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} e^{A_s h} + A_s^{-1}(e^{A_s(h-\tau)} - I)B_s K C_s & e^{A_s(h-\tau)} A_s^{-1}(e^{A_s \tau} - I)B_s K \\ C_s & 0 \end{bmatrix}$$



from simulations $\Rightarrow$ for $h=1.2$
you have instability
when $\tau=0.8$



Outline

- Reference configuration
- Examples (constant delay)
- **Examples (time-varying sampling time and delay)**
- Compensation of delays
- Examples (constant sampling time and delay, with packet dropouts)

- What happens when h, τ are time varying system?



time varying parameters case:

CASE 2 h_k, τ_k time-varying

considering dynamic changes over time

$$z_{k+1} = \underbrace{\Phi(h_k, \tau_k)}_{\text{depends on } \Phi, \text{ still on its eigenvalues?}} z_k$$

not enough to compute
eig(Φ) to analyze stability

depends on Φ ,
still on its eigenvalues?

↓
← NO

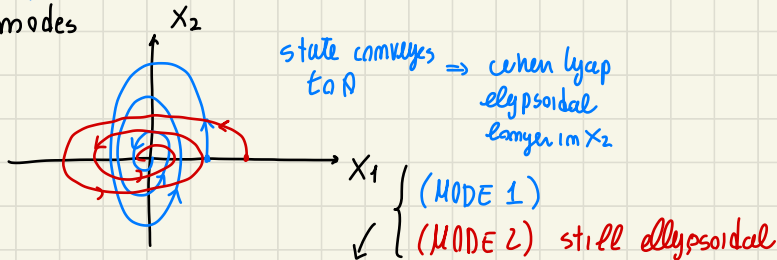
$|\text{eig}(\Phi)| < 1 \forall \tau_k, h_k$ NON CNS! neither CN OR CS!!

"Shur stability" of $\Phi(h_k, \tau_k) \forall h_k, \tau_k$ is neither necessary nor sufficient for stability of NCS

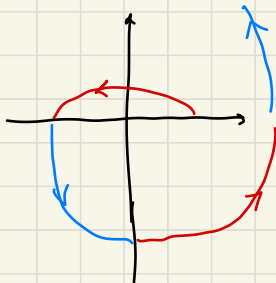
↓ let's analyze a simple switching mode with 2 modes both A.S

example

2 modes



the switching between the two induce instability, because different $\perp$ Lyap



IF you switch

in that cases,

alone both A.S

BUT if switch @ wrong time $\Rightarrow$ unstable system!

(OR it can happen opposite 2 unstable switch $\Rightarrow$ stable)

I cannot rely on eigs $\rightarrow$ shape of Lyapunov function
and switching time affect stability!

this can happen for example

$$\begin{array}{l} x_{k+1} = F_1 x_k \quad (\text{mode 1}) \\ x_{k+1} = F_2 x_k \quad (\text{mode 2}) \end{array} \quad \updownarrow \quad \text{switching @ each } k \text{ time step}$$

$\Downarrow$
you see the 2 step evolution

$$x_{k+2} = \boxed{F_1 F_2} x_k \quad \text{"time-invariant" evolution}$$

so we can just
check $F_1 F_2$ eigenvalues

$$\text{eig}(F_1 F_2) \neq \text{eig}(F_1) \text{eig}(F_2)!$$

$$\text{eig}(F_1 F_2) = \text{eig}(F_2 F_1)$$

$$x_{k+2} = \boxed{F_2 F_1} x_k \quad \text{if opposite switch}$$

$\uparrow$ different matrix BUT
with same eigenvalues

$$F_1 = \begin{bmatrix} 0.7384 & 0.5233 \\ 0.9764 & 0.4299 \end{bmatrix} \quad F_2 = \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}$$

unstable $\rightarrow$ stable $\Rightarrow$ switching stable!

OR

switching from stable $\rightarrow$ unstable $\Rightarrow$ unstable!

OR

stable $\rightarrow$ stable $\Rightarrow$ stable

OR

stable $\rightarrow$ stable $\Rightarrow$ unstable

OR

unstable $\rightarrow$ unstable $\Rightarrow$ stable

unstable $\rightarrow$ unstable $\Rightarrow$ unstable

$\nwarrow$ all can happen!
all comb.
are possible

eg. values alone tell nothing about overall stability property!

↓
how to study stability? → luckily we know how to use LM



LM help to solve this problem!

We had 2 Lyapunov function different (shapes) the switching between the two modes Lyap lead to different behavior

↓
find common Lyap for all matrix ⇒ to guarantee overall traj

(common traj by same dynamic)



Theorem

The NCS is asymptotically stable IF

$$\exists P > 0 \text{ s.t. } \Phi(l, \tau)^T P \Phi(l, \tau) - P < 0$$

assumptions l_k, τ_k t-varying

$$l_{\min} \leq l_k \leq l_{\max}$$

$$\tau_{\min} \leq \tau_k \leq \tau_{\max}$$

= 0 ideally

$\forall \tau, l$

similar to previous condition, now must hold $\forall \tau, l$

← same P!

(more conservative than just eigenvalues, same Lyap for more dynamic!)

← $\forall$ dynamics same P, using same Lyap. function & dym.



intuitively, we can prove it by Lyap. theory all dym has Lyap:

$$V(z_k) = z_k^T P z_k$$

$$\Rightarrow V(z_{k+1}) = z_{k+1}^T P z_{k+1} = (\Phi(l_k, \tau_k) z_k)^T P \Phi(l_k, \tau_k) z_k =$$

$$= z_k^T \underbrace{\Phi(h_k, z_k)^T P \Phi(h_k, z_k)}_{< P \text{ always from hyp!}} z_k$$

so for sure $V(z_{k+1}) < z_k^T P z_k = V(z_k)$ so for Lyap. theorem state converges to 0!

this cannot be

done with usual tool, you need LMI to do this!

↓
example range of application
of your analysis theory

ONLY

SUFFICIENT

(not necessary condition) → it is particular case for which $P = \text{const}$

- much more difficult to find P_k
 - so only sufficient,
even if you don't
find it P , maybe asymp. stable $\Rightarrow$ you constrain P
to be constant
- ↓ easy analytical condition